

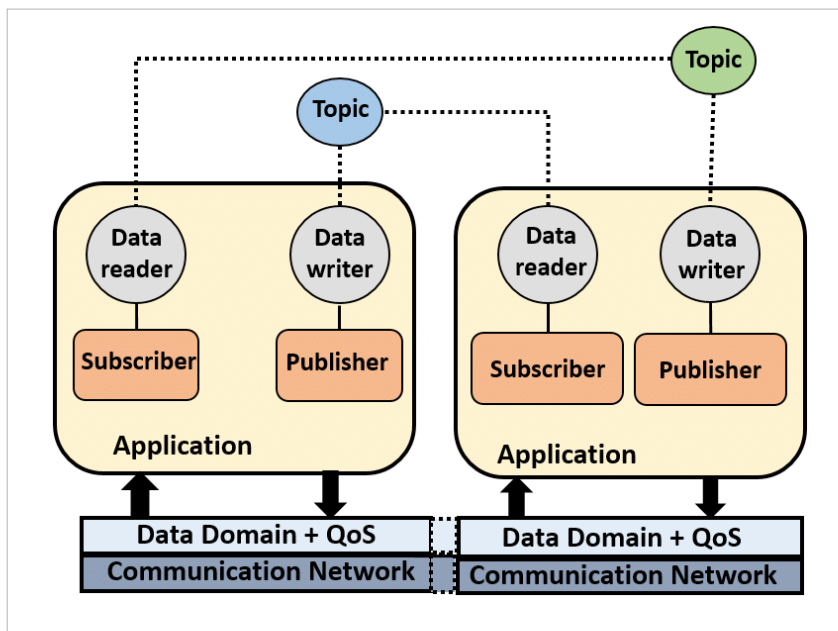


Figure 5: DDS data transaction model (Courtesy: ResearchGate)

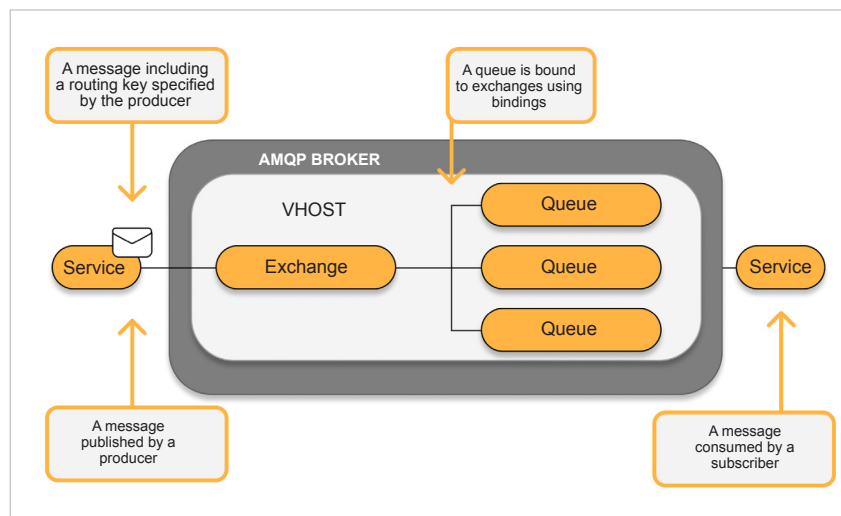


Figure 6: AMQP data transaction model (Courtesy: CloudAMQP)

As shown in Figure 5, the DDS architecture consists of a data-centric publish/subscribe layer (DCPL) in addition to an optional data-local reconstruction layer (DLRL). Scalability, resource awareness and fast paced distribution of data are ensured with the help of the DCPL layer, while the DLRL layer provides an interface for the operations of the DCPL layer.

Advanced message queuing protocol (AMQP)

AMQP is another publish/subscribe model based protocol that is primarily focused on guaranteed completion of transactions. Designed back in 2003, it was used mostly in the financial sector for obvious reasons. AMQP makes use of advanced queuing and distribution techniques to ensure security, reliability and speed.

As seen in Figure 6, the publisher publishes the message to the AMQP broker with a predesignated routing key attached to the message. Utilising the routing key, the broker queues the message for delivery to the designated consumer. These messages are bound to exchanges with the help of bindings. There are, however, a few drawbacks of the AMQP model which has restricted its mass adoption in the IoT space. For one, it is a heavy protocol that is not ideal for smaller IoT devices with a lack of bandwidth and power. Second, the guaranteed delivery of messages is not always a high priority for all IoT projects.

Lightweight M2M (LwM2M)

Designed by the Open Mobile Alliance (OMA) in 2014, LwM2M has been specifically designed to tackle the challenges of modern day IoT systems. With over 75 billion working IoT devices estimated across the world by 2025, LwM2M has been designed for scalability, speed and reliability in a constrained environment.

As shown in Figure 7, the LwM2M server stack consists of all the essential components that an IoT network requires including CoAP protocol, UDP protocol, SMS bearer and an efficient payload system. It also has DTLS security inbuilt. Additionally, LwM2M provides a device level bootstrapping facility, which gives you the ability to customise the usage of the protocol as per your requirements on a local level. It also provides a client registration interface, and access to resources and instances. This gives the user access to customise the settings on a granular level without having to revert to factory settings.

Which IoT protocol is right for your project?

What is interesting about IoT, unlike other tech projects, is that the requirements vary a lot from